

LM in Artificial Intelligence

Image Processing and Computer Vision

(Example of Exam)

- 1) List the main steps of the DoG keypoint detection algorithm, providing a very brief description for each step. Report the equation to define sigma for each scale in an octave. Describe the motivations behind exploiting multiple octaves and how they are computed.
- 2) Discuss the limitations of fully connected linear layers when used to process $H \times W$ grayscale images. Then discuss the inductive biases of convolutions and how they help to overcome the previous limitations. Finally, report the expression to compute each entry of the output activation, clearly defining all the terms used in it.